YUANKAI TAO

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EDUCATION

Georgia Institute of Technology *Aug 2026 — Dec 2027*
M.S. in Operations Research, Stewart School of Industrial and Systems Engineering
Selected Coursework: Deterministic Optimization, Computational Statistics, Simulation

University of California, San Diego *Sep 2024 — Dec 2025*
M.S. in Quantitative Finance, Rady School of Management *GPA:* 3.84/4.0
Selected Coursework: Investment Analysis, Fiancial Data Collection & Analysis, Financial Statement Analysis

Zhongnan University of Economics and Law *Sep 2020 — Jun 2024*
B.S. in Management(ACCA track), School of Accounting and Finance *GPA:* 3.63/4.0
Selected Coursework: Financial Reporting, Audit & Assurance, Probability & Statistics, Python Programming

TECHNICAL SKILLS

Programming & Tools Excel, Word, PowerPoint, SQL, Python, Scikit-learn, PyTorch
Quantitative Methods Financial Modeling, Statistical Analysis, Data Visualization, Machine Learning

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd. *May 2026 — Jul 2026*
Research Analyst, Intern *Quantitative Strategy Department | Hong Kong*

• Analyzed copper futures price trends and volatility patterns using historical market data, technical indicators, and event-driven analysis to assess market dynamics and potential investment risks.

• Built a dynamic VaR/CVaR risk measurement system for copper futures; introduced semi-variance slicing method to isolate long/short momentum risks; constructed and backtested 17 risk models combining 5 volatility models (GARCH, EWMAV) with 3 distribution assumptions (Normal, StudentSkewed-t, Filtered Historical Simulation).

• Developed a monthly rebalanced cross-sectional futures strategy using LightGBM return probabilities with 120-day rolling retraining window; implemented in Backtrader achieving 1.89 Sharpe ratio after transaction costs and 15.51% excess return comparing to the benchmark.

Haitong Securities *Jan 2024 — May 2024*
Investment Analyst, Intern *Investment Banking Department*

• Conducted due diligence using financial terminals, corporate filings, and industry reports; assessed client creditworthiness and identified potential financial and operational risks.

• Analyzed historical sales and procurement data, investigated material fluctuations and unusual transactions, and evaluated underlying business drivers to support risk assessment and IPO review.

• Conducted comparative analysis of R&D spending and key operating expenses across industry peers; identified cost structure differences and summarized variance drivers.

Ernst & Young *Dec 2022 — Feb 2023*
Audit, Intern *Financial Service Office (FSO)*

• Analyzed intercompany and investment transactions for Hyundai Motor Finance Company; performed variance analysis, investigated significant deviations, reviewed supporting records, and documented findings and recommendations.

• Automated the cleaning and consolidation of financial tables from 220+ CSV files, producing standardized audit outputs and improving the efficiency and consistency of recurring reporting.

SELECTED PROJECTS

Kaggle - Jane Street Market Prediction Competition *Silver Medal*
• Built a memory-efficient Polars/PyTorch pipeline for 47M intraday anonymized market observations, engineering lagged, rolling and temporal features.

• Contributed to the baseline training and final ensemble of XGBoost, MLP, Ridge, and TabM, increasing the generalization performance of the model sets over fine-tuning, helping the team finish 94th among 3,757 teams.

Text Analysis and BERT Fine-tuning on SEC 10-K Reports *MGTF 423 Final Project, UCSD*

• Conducted exploratory NER and frequency analysis on Fluence Energy Inc. SEC filings.

• Implemented iterative human-in-the-loop fine-tuning with 3 rounds of annotation; trained BERT model achieving 79% accuracy in quantifying uncertainty tone in financial disclosures.

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill *Mar 2025 — Jan 2026*

The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

• Engineered large-scale data pipeline (Python, PostgreSQL, PostGIS) processing 76-million-observation spatial dataset on Google Cloud Platform; managed 964.5M-record SQL database with 100GB+ data on UNC Longleaf HPC cluster using Python and DuckDB.

• Executed complex geospatial queries to extract population statistics with property-related constraints.

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate) — Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate) — Coursera, UC San Diego

Beta Gamma Sigma (Top 20% of graduating class) — UCSD, *2024 — 2025*

Special Scholarship in Innovation and Entrepreneurship — ZUEL, *2022 — 2023*